3.3 Elements

Matter is classified into two broad categories, namely, pure substances and mixtures. Pure substances are further divided into two categories as elements and compounds.

Matter in its simplest form is called an element. We are using many elements in our daily life. The common salt consists of two elements, sodium and chlorine. Water consists of hydrogen and oxygen. Magnesium and phosphorus are used for making crackers. Sulphur is used as manure in agriculture. Gallium is used for making mobile phones and silicon is used for making computer chips.

There are 118 known elements till date. Out of these, 94 elements occur naturally while 24 elements are synthesised artificially in the laboratory.

3.3.1 Classification of Elements

We can classify the elements broadly into metals, non-metals and metalloids based on their chemical properties.



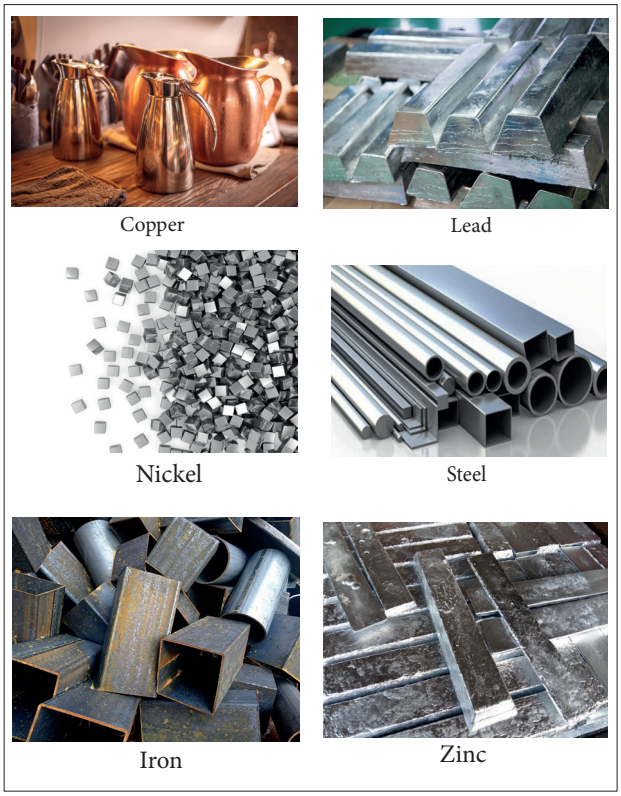
Robert Boyle is the first scientist who used the term element. He is the early proponent of the elemental nature of matter and the nature of vacuum. He is known best for Boyle's Law.



Metals

We have tools, utensils and jewellery made of silver, copper, iron, gold, aluminium, etc. By hammering or rolling we can deform these materials into various shapes. Such elements that are malleable (a material may be flattened into thin sheets or various shapes) are called as metals.

Metals are generally hard and shiny elements. Sodium is one of the exceptions as it is soft. All metals, except mercury are solids at room temperature. Mercury is the only metal that is liquid at room temperature. Metals are malleable, can be bent or beaten into sheets. They can be drawn into wires. They are good conductors of heat and electricity. Copper, lead, tin, nickel, iron, zinc, gold, magnesium and calcium are examples of metals.



Metals

Non-metals

Non-metals are generally dull and soft. However, diamond is shiny and also the hardest natural substance on earth. Non-metals can be gases, solids and liquids. Non-metals such as oxygen, hydrogen and chlorine are gases at room temperature. Carbon, iodine, sulphur and phosphorus are solids at room temperature. Bromine is the only non-metal that is liquid at room temperature. Non-metals are poor conductors of heat and electricity. However, graphite (a form of the non-metal carbon) is a good conductor of electricity.